BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Array Traversal and Updates

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is a one-dimensional array?

2. Valid ages are 13-18. Which is normal test data?

A) 12 B) 15
C) 19 D) 'twelve'

3. "Any device connected to the Internet is automatically an embedded system." Correct this. Explain why.

4. A pupil enters a domain name. "DNS gives the same MAC address to every visitor." Correct this. How does DNS help find the website?

Today's objectives

Use loops to process and update array values. Use the technique in an independent scenario.

Key vocabulary - Read, cover, recall

Traversal
A loop visits every array element using a valid index.

Accumulator
A total variable stores the running sum and is updated for each value.

FOR iteration
A counted loop can process indices zero through length minus one.

Average
Total divided by number of elements gives the mean.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Traversal	A loop visits every array element using a valid index. Apply this in a competition points.
Accumulator	A total variable stores the running sum and is updated for each value. Apply this in a four temperatures.
FOR iteration	A counted loop can process indices zero through length minus one. Apply this in a café takings.
Average	Total divided by number of elements gives the mean. Apply this in a weekly marks.

I Do - One worked example

A competition points is being developed. Explain how traversal and accumulator are used. [4 marks]

1. Read the scenario and identify the data the program needs.

2. Traversal [1]; A loop visits every array element using a valid index. [1].

3. Accumulator [1]; A total variable stores the running sum and is updated for each value. [1].

4. Two linked points for each concept.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A four temperatures uses traversal. Explain how it helps. [2 marks]

2. Explain accumulator in a café takings. [2 marks]

3. A weekly marks uses for iteration and average. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the line that adds each file size to the total.

```
array imageMB = [2,5,3]
total = 0
for i = 0 to 2
// add imageMB[i]
next i
print(total)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO relevant techniques from this topic. [2 marks]

Q3. Explain how accumulator and for iteration help with a fitness step counts. [4 marks]

Q4. A programmer designs a lunch orders. Explain why traversal, for iteration and average matter. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A loop visits every array element using a valid index. ☐ T ☐ F
2. A loop visits every array element using a valid index ☐ T ☐ F
3. A total variable stores the running sum and is updated for each value ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 A fixed-size indexed collection of related elements; OCR ERL indexing begins at zero. ☐

Q2 B - 15 ☐

Q3 It is part of a heating system, with the specific task of controlling temperature. ☐

Q4 DNS servers look up the IP address for a domain name. Access needs the relevant server and network services to work. ☐

Code First - Model

```
array imageMB = [2,5,3]
total = 0
for i = 0 to 2
total = total + imageMB[i]
next i
print(total)
```

Output: 10

We Do - Check the model

1. Traversal [1]; A loop visits every array element using a valid index. [1]. ☐

2. Accumulator [1]; A total variable stores the running sum and is updated for each value. [1]. ☐

3. FOR iteration [1]; A counted loop can process indices zero through length minus one. [1]. Average [1]; Total divided by number of elements gives the mean. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Traversal [1]; FOR iteration [1]. ☐

Q3 Accumulator [1]; A total variable stores the running sum and is updated for each value. [1]. FOR iteration [1]; A counted loop can process indices zero through length minus one. [1]. ☐

Q4 Traversal [1]; A loop visits every array element using a valid index. [1]. FOR iteration [1]; A counted loop can process indices zero through length minus one. [1]. Average [1]; Total divided by number of elements gives the mean. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: